Introduction to Machine Learning

A Small Practice Document for Your RAG Assistant

This document is a beginner-friendly sample knowledge base. You can upload it to your RAG application, ask questions about it, and check whether the system retrieves relevant information.

1. What is Machine Learning?

Machine learning is a branch of artificial intelligence that enables computer systems to learn patterns from data and use those patterns to make predictions or decisions. Instead of writing a separate rule for every possible situation, developers train a model using examples.

2. Types of Machine Learning

Supervised learning: The model learns from labelled examples. Each training example contains input data and a known target. Classification and regression are common supervised-learning tasks.

Unsupervised learning: The model receives data without target labels and attempts to discover structure. Clustering is a common example; it groups similar data points together.

Reinforcement learning: An agent interacts with an environment, takes actions, and receives rewards or penalties. The agent learns a policy that aims to maximize cumulative reward.

3. Training and Testing

A dataset is often divided into training, validation, and test sets. The training set is used to fit model parameters. The validation set helps compare configurations and tune hyperparameters. The test set is held back until final evaluation to estimate performance on unseen examples.

4. Overfitting and Underfitting

Overfitting occurs when a model learns details and noise in the training data so closely that it performs poorly on new data. Underfitting occurs when a model is too simple to capture important patterns, resulting in poor performance on both training and unseen data.

5. Classification and Regression

Classification predicts a category, such as whether an email is spam or not spam. Regression predicts a numerical value, such as the price of a house. The choice of task depends on the type of outcome being predicted.

6. Model Evaluation

Accuracy is the fraction of predictions that are correct. Precision measures how many predicted positive cases are truly positive. Recall measures how many actual positive cases the model identifies. The F1-score is the harmonic mean of precision and recall. For regression, mean absolute error and mean squared error are commonly used.

7. Why Data Quality Matters

A machine-learning model depends on the examples used to train it. Missing values, incorrect labels, duplicated records, and biased sampling can reduce model quality. Data cleaning and careful evaluation are important parts of a reliable machine-learning workflow.

8. Example Questions for Testing RAG

1. What is machine learning?
2. What is the difference between supervised and unsupervised learning?
3. Why do we keep a test set separate?
4. Explain overfitting in simple words.
5. What is the difference between classification and regression?
6. Which metrics are used to evaluate a classifier?